Section 1.0: Document Control & Engineering Metadata

Document Control Information:
Document ID: HLD-PT-2026-001-REV1
System Domain: Powertrain Domain Controller (ECU_EMS_01)
Architecture Standard: AUTOSAR Classic Platform Release 4.4.0
Target Microcontroller: Infineon AURIX TC397XX (Tri-Core 300MHz Lockstep)
Operating System: AUTOSAR OS Scalability Class 4 (SC4)
Compiler: Tasking VX-toolset for TriCore v6.3r1

Revision History:
- Version 0.1 (Jan 2026): Initial Architectural Draft and Allocation
- Version 0.5 (May 2026): Component Interface and RTE Runnable Mapping
- Version 1.0 (Sep 2026): Approved Baseline Architecture for Firmware Release 1.0

Stakeholder Sign-Offs:
- Lead System Architect: Dr. E. Schmidt (Approved)
- Functional Safety Manager: H. Weber (ASIL B Qualified)
- Integration Lead: K. Patel (Approved)

Scope and Document Organization:
This High-Level Design document establishes the formal software architecture decomposition for the Powertrain Domain Controller ECU. It defines the software components, port interfaces, runnable entities, task scheduling, CAN FD network matrices, and diagnostic event reporting.

Section 2.0: Executive System Overview & ECU Topology

2.1 Executive Powertrain Domain Scope:
The Electric Powertrain Domain Controller serves as the central supervisory computation unit for electric vehicle torque delivery, driver pedal interpretation, cruise control torque limiting, traction motor interface coordination, and high-voltage safety monitoring.

2.2 Hardware Topology & Vehicle Network Context:
The ECU connects to the vehicle communication backbone via two independent bus systems:
1. Powertrain CAN FD Channel A (500 kbps nominal, 2.0 Mbps data phase) for drivetrain commands.
2. Diagnostic & Calibration CAN FD Channel B (500 kbps / 2.0 Mbps) for OBD/UDS diagnostics.
3. Private SPI Interface connecting to the Inverter Gate Driver Complex Device Driver (CDD).

2.3 Computational Hardware Allocation:
- Core 0: Basic Software Stack (BSW), Communication Drivers (CanIf, PduR), OS Scheduler.
- Core 1: High-priority torque arbitration algorithms and sensor processing (Lockstep Core).
- Core 2: Diagnostic Event Manager (DEM), UDS Server (DCM), and Non-Volatile Memory (NvM).

Section 3.0: AUTOSAR Classic Architectural Layering

3.1 Layered Architecture Overview:
In accordance with AUTOSAR Classic specification 4.4.0, the software is divided into:
- Application Layer (SWCs): Vehicle control algorithms independent of underlying microcontroller.
- Runtime Environment (RTE): Abstract communication middleware connecting SWC ports.
- Basic Software (BSW): System services, memory stack, diagnostic stack, and comms.
- Microcontroller Abstraction Layer (MCAL): Hardware-specific peripheral drivers.

3.2 Inter-Runnable Communication Paradigms:
1. Implicit Data Communication: Used for high-frequency periodic control loops to prevent jitter.
2. Explicit Data Communication: Used for event-driven status flags and mode requests.
3. Client-Server Invocations: Synchronous and asynchronous function calls for diagnostics.

3.3 BSW Module Allocation:
- System Services: EcuM (ECU State Manager), BswM (BSW Mode Manager), OS (Operating System).
- Memory Services: NvM (Non-Volatile RAM Manager), Fee (Flash EEPROM Emulation).
- Communication Services: Com (COM Manager), PduR (PDU Router), CanIf (CAN Interface).

Section 4.0: Functional Safety Concept (ISO 26262)

4.1 Item Definition & Hazard Analysis (HARA):
Under ISO 26262:2018, the Powertrain Domain Controller has been assessed against hazardous events:
- Hazard H-01: Unintended vehicle acceleration due to torque over-estimation -> Safety Goal SG-01 (ASIL B).
- Hazard H-02: Loss of regenerative braking capability without driver warning -> Safety Goal SG-02 (ASIL B).
- Hazard H-03: Throttle valve stuck in open position -> Safety Goal SG-03 (ASIL B).

4.2 ASIL Decompositions & Classifications:
- TorqueArbitrationSWC: Allocated ASIL B (SG-01 implementation).
- EngineSpeedSensorSWC: Allocated ASIL B (Sensor acquisition plausibility).
- ThrottleActuatorSWC: Allocated ASIL B (Direct actuator output control).
- DiagnosticEventManagerSWC: Allocated QM (Quality Management - non-safety critical).
- Memory Management Stack (NvM): Allocated QM with CRC integrity verification.

4.3 Freedom From Interference (FFI) Mechanisms:
Memory Protection Unit (MPU) regions are configured on the Aurix TC397 to partition ASIL B
memory buffers from QM diagnostic memory, ensuring QM tasks cannot corrupt torque tables.

Section 5.0: Application Layer Software Component Architecture

5.1 Software Component Taxonomy:
The application software consists of distinct AUTOSAR SWC archetypes:
1. Application SWCs: Core vehicle logic components containing control algorithms.
2. Sensor-Actuator SWCs: Device-oriented components abstracting physical inputs/outputs.
3. Service SWCs: Interfaces to Basic Software services (DEM, NvM, ComM).
4. Complex Device Drivers (CDD): Direct hardware access for timing-critical SPI drivers.

5.2 Primary Component Inventory:
- TorqueArbitrationSWC: Application SWC (Arbitration of driver pedal and cruise demand).
- EngineSpeedSensorSWC: Sensor Actuator SWC (Crankshaft Hall sensor acquisition).
- ThrottleActuatorSWC: Sensor Actuator SWC (Electronic throttle H-bridge drive).
- CruiseControlSWC: Application SWC (Velocity regulation and speed limit enforcement).
- ThermalManagementSWC: Application SWC (Inverter temperature and coolant pump control).
- HighVoltageMonitorSWC: Sensor Actuator SWC (High-voltage DC bus voltage and current).

Section 6.0: TorqueArbitrationSWC Architectural Specification

6.1 Component Purpose:
TorqueArbitrationSWC arbitrates multiple concurrent torque requests:
1. Driver pedal demand (accelerator pedal position sensor).
2. Cruise control speed regulation torque request.
3. Traction control system (TCS) wheel slip limiting torque.

6.2 Execution Scheduling:
- Periodic Rate: 10ms (mapped to OsTask_10ms_Core1, Task Priority 15).
- Worst-Case Execution Time (WCET): 1.82ms on AURIX Core 1.
- ASIL Allocation: ASIL B.

6.3 Port Interfaces:
- Port Rp_DriverDemand (Required): Interface If_DriverDemand (SenderReceiverInterface).
  * Data Element AccelPedalPosition_pct (uint8, 0..100%).
  * Data Element KickdownSwitch (boolean).
- Port Rp_EngineSpeed (Required): Interface If_EngineSpeed (SenderReceiverInterface).
  * Data Element EngineSpeed_rpm (uint16, 0..8000 RPM).
- Port Pp_TorqueCommand (Provided): Interface If_TorqueCommand (SenderReceiverInterface).
  * Data Element TargetTorque_Nm (sint16, -500..+1200 Nm).
  * Data Element TorqueValid (boolean).

Section 7.0: EngineSpeedSensorSWC Architectural Specification

7.1 Component Purpose:
EngineSpeedSensorSWC interfaces with the crankshaft hall-effect sensor via a Complex Device Driver (CDD_Crankshaft). It provides rotational speed calculation, tooth pulse verification, and rate-of-change filtering.

7.2 Execution Scheduling:
- Periodic Rate: 10ms (mapped to OsTask_10ms_Core1).
- ASIL Level: ASIL B.

7.3 Ports and Interfaces:
- Port Pp_EngineSpeed (Provided / P-Port):
  * Interface: If_EngineSpeed (SenderReceiverInterface)
  * Data Elements:
    - EngineSpeed_rpm (uint16, Resolution: 0.25 RPM/bit, Range: 0 to 8000 RPM).
    - SpeedSignalStatus (uint8: 0x00=VALID, 0x01=INITIALIZING, 0x02=FAULT).
- Port Pp_SensorDiagnostics (Provided / P-Port):
  * Interface: If_SensorDiag (ClientServerInterface)
  * Operations: GetSensorHealthStatus(out uint8 status).

Section 8.0: ThrottleActuatorSWC Architectural Specification

8.1 Component Purpose:
ThrottleActuatorSWC translates arbitrated torque demands into electronic throttle body valve positions and drives the PWM H-bridge power driver.

8.2 Execution Scheduling:
- Periodic Rate: 10ms (mapped to OsTask_10ms_Core1).
- ASIL Level: ASIL B.

8.3 Port Interfaces:
- Port Rp_TorqueCommand (Required / R-Port):
  * Interface: If_TorqueCommand (SenderReceiverInterface).
  * Data Element: TargetTorque_Nm (sint16).
- Port Pp_ActuatorState (Provided / P-Port):
  * Interface: If_ActuatorFeedback (SenderReceiverInterface).
  * Data Element: ThrottleAngle_deg (uint16, Range: 0..900, 0.1 deg/bit).
- Port Pp_ActuatorFault (Provided / P-Port):
  * Interface: If_ActuatorFaultReport (ClientServerInterface).

8.4 Actuator Fail-Safe Behavior:
Upon receiving TorqueValid == FALSE or communication timeout, the actuator defaults to a mechanical spring-return limp-home throttle opening angle of 7.5 degrees.

Section 9.0: CruiseControlSWC Architectural Specification

9.1 Component Purpose:
CruiseControlSWC implements vehicle speed maintenance and driver-selected target speed control.

9.2 Execution Scheduling:
- Periodic Rate: 20ms (mapped to OsTask_20ms_Core1).
- ASIL Level: ASIL B.

9.3 Port Interfaces:
- Port Rp_DriverCruiseSwitches (Required / R-Port):
  * Interface: If_CruiseSwitches (SenderReceiverInterface).
  * Data Elements: SetSpeed_kmh (uint8), CruiseActive (boolean), CancelPressed (boolean).
- Port Pp_CruiseTorqueLimit (Provided / P-Port):
  * Interface: If_CruiseTorque (SenderReceiverInterface).
  * Data Elements: MaxAllowedTorque_Nm (uint16), SpeedError_kmh (sint8).

9.4 Safety Interlocks:
Cruise control automatically disengages within 15ms upon detection of brake pedal depression
or ESC intervention request.

Section 10.0: ThermalManagementSWC Architectural Specification

10.1 Component Purpose:
ThermalManagementSWC monitors inverter IGBT junction temperatures, motor stator temperatures, and commands electric coolant circulation pump duty cycles.

10.2 Execution Scheduling:
- Periodic Rate: 100ms (mapped to OsTask_100ms_Core2).
- ASIL Level: QM (Quality Management).

10.3 Port Interfaces:
- Port Rp_InverterTemps (Required / R-Port):
  * Interface: If_ThermalSensors (SenderReceiverInterface).
  * Data Elements: TempIGBT_degC (sint16), TempMotor_degC (sint16).
- Port Pp_PumpCommand (Provided / P-Port):
  * Interface: If_CoolantPump (SenderReceiverInterface).
  * Data Elements: PumpDutyCycle_pct (uint8, 0..100%).

10.4 Overtemperature Derating Logic:
When TempIGBT_degC exceeds 115°C, ThermalManagementSWC publishes a DerateTorque_pct signal demanding linear torque reduction down to 40% at 130°C.

Section 11.0: High-Voltage Battery Interface Architecture

11.1 Component Purpose:
HighVoltageMonitorSWC interfaces with the external Battery Management System (BMS) over CAN FD. It reads available discharge power limits, pack voltage, and High Voltage Interlock (HVIL) status.

11.2 Execution Scheduling:
- Periodic Rate: 20ms (mapped to OsTask_20ms_Core0).
- ASIL Level: ASIL B.

11.3 Port Interfaces:
- Port Rp_BMS_Limits (Required / R-Port):
  * Interface: If_BatteryPowerLimits (SenderReceiverInterface).
  * Data Elements: MaxDischargePower_kW (uint16), BatteryVoltage_V (uint16).
- Port Pp_HVIL_Status (Provided / P-Port):
  * Interface: If_HVIL_State (SenderReceiverInterface).
  * Data Elements: HVIL_Closed (boolean).

11.4 Emergency Power Shutdown:
If HVIL_Closed transitions to FALSE while vehicle speed > 0, an immediate emergency inverter discharge sequence is triggered via Dem Event DEM_EVENT_HVIL_OPEN.

Section 12.0: Non-Volatile Memory (NvM) Stack Architecture

12.1 NvM Block Allocation Matrix:
Non-Volatile Memory stores calibration offsets, learned throttle positions, and DTC fault records:
- Block NvM_Block_ThrottleCalibration:
  * Size: 64 Bytes, Management: NVRAM_BLOCK_DATASET, CRC: CRC16.
  * Restored at ECU startup (EcuM Startup Phase II).
- Block NvM_Block_DTC_FaultMemory:
  * Size: 512 Bytes, Management: NVRAM_BLOCK_REDUNDANT (Double-stored with inverted copy).
  * Target: Internal Emulated Flash (Fee).

12.2 Fee & Fls Integration:
Flash EEPROM Emulation (Fee) provides virtual 32-bit block addressing over internal Flash sectors with automatic wear-leveling and background garbage collection.

Section 13.0: CAN FD Bus Matrix & Frame Communication

13.1 CAN Controller Configuration:
- Controller: Aurix MultiCAN+ Module CAN0.
- Baudrate Nominal: 500 kbps (Sample Point: 80%).
- Baudrate Data Phase: 2000 kbps (2.0 Mbps, Sample Point: 75%).
- Transceiver: NXP TJA1044GT with standby mode control.

13.2 Transmitted Frames (Tx):
- Frame CAN_Powertrain_TorqueDemand:
  * Frame ID: 0x1A4 (Standard 11-bit ID).
  * Cycle Time: 10ms periodic.
  * Payload Length: 16 Bytes (CAN FD DLC 16).
  * Signal TargetTorque_Nm: Byte 0-1 (sint16, Intel Endian).
  * Signal E2E_Counter: Byte 7 (bits 0-3).
  * Signal E2E_CRC8: Byte 8 (CRC8 0x2F).

- Frame CAN_EngineSpeed_Broadcast:
  * Frame ID: 0x2B0, Cycle Time: 20ms, Payload Length: 8 Bytes.

Section 14.0: End-to-End (E2E) Communication Protection

14.1 E2E Protection Concept:
To satisfy ISO 26262 ASIL B data integrity requirements on bus communications, E2E Library Profile P01 is integrated into the PDU Router (PduR) and COM stack.

14.2 E2E Profile P01 Specification:
- Protected Message: CAN_Powertrain_TorqueDemand (ID: 0x1A4).
- Data ID: 0x12 (Unique 16-bit identifier preventing masquerading).
- Counter: 4-bit alive counter (0 to 14, incremented every transmission cycle).
- Checksum: 8-bit CRC computed using polynomial 0x1D (SAE-J1850).
- Timeout Monitoring: Maximum permissible reception delay = 35ms (3 missed frames).

14.3 Error Handling on E2E Failure:
If E2E validation returns E2E_E_WRONGCRC or E2E_E_WRONGCOUNTER for 3 consecutive frames, the receiver marks signal status as INVALID and falls back to safe limp-home mode.

Section 15.0: Diagnostic Stack (DEM & UDS ISO 14229) Architecture

15.1 Diagnostic Stack Overview:
The diagnostic subsystem comprises:
- DCM (Diagnostic Communication Manager): Handles UDS ISO 14229-1 request parsing.
- DEM (Diagnostic Event Manager): Manages fault memory, DTC debounce, and freeze frames.
- FIM (Function Inhibition Manager): Inhibits software functions based on active DTCs.

15.2 Supported UDS Diagnostic Services:
- Service 0x10: DiagnosticSessionControl (Default, Extended, Programming Sessions).
- Service 0x11: ECUReset (HardReset, SoftReset).
- Service 0x14: ClearDiagnosticInformation.
- Service 0x19: ReadDTCInformation (ReportDTCByStatusMask, ReportSnapshotData).
- Service 0x22: ReadDataByIdentifier (Read live sensor signals and calibrations).
- Service 0x2E: WriteDataByIdentifier (Configuration writing with security access).
- Service 0x27: SecurityAccess (Seed & Key cryptography for seed length 64 bits).

Section 16.0: UDS Data Identifier (DID) Master Catalogue

16.1 Supported Diagnostic Data Identifiers (Service 0x22):
- DID 0xF190: Vehicle Identification Number (VIN)
  * Format: ASCII, Length: 17 Bytes, Access: Read-Only (All Sessions).
- DID 0xF188: Vehicle Manufacturer ECU Software Number
  * Value: 'SW-EMS-TC397-V1.0', Length: 16 Bytes.
- DID 0x2100: Current Arbitrated Torque Output
  * Payload: TargetTorque_Nm (sint16), Unit: 0.1 Nm/bit, Length: 2 Bytes.
- DID 0x2105: Accelerator Pedal Position Live Sensor Value
  * Payload: AccelPedal_Pct (uint8, 0..100%), Length: 1 Byte.
- DID 0x2110: Engine Rotational Speed (Crankshaft Sensor)
  * Payload: EngineSpeed_rpm (uint16, 0.25 rpm/bit), Length: 2 Bytes.
- DID 0x2120: Throttle Valve Angle Actual Position
  * Payload: ThrottleAngle_deg (uint16, 0.1 deg/bit), Length: 2 Bytes.

Section 17.0: Diagnostic Trouble Codes (DTC) & Fault Debounce

17.1 DTC Master Table:
- DTC 0x921401: Throttle Position Sensor Circuit Voltage Above Threshold
  * Fault Symptom: Accelerator pedal / throttle angle tracking error > 12%.
  * Debounce Strategy: Pre-debounce counter (Step Up: 2, Step Down: 1, Threshold: 20).
  * Aging Threshold: 40 warm-up cycles without occurrence.
  * Fail-Safe Action: Torque limit clamped to max 50 Nm; MIL lamp illuminated.

- DTC 0x921502: Engine Speed Signal Plausibility Fault
  * Fault Symptom: Measured rate of engine speed change exceeds physically possible RPM/sec.
  * Debounce Strategy: Time-based debounce (Timer: 150ms).
  * Fail-Safe Action: Switch to secondary wheel-speed derived estimate.

- DTC 0x922004: CAN_Powertrain_TorqueDemand E2E Protection Failure
  * Debounce Strategy: Consecutive counter failure (3 frames).

Section 18.0: Operating System (OS) Task & Core Mapping

18.1 Task Scheduling Matrix:
AUTOSAR OS priority-driven preemptive scheduling table:
- Task OsTask_10ms_Core1 (Priority 15, Core 1):
  * TorqueArbitrationSWC_MainFunction (Exec Time: 1.8ms)
  * EngineSpeedSensorSWC_MainFunction (Exec Time: 0.6ms)
  * ThrottleActuatorSWC_MainFunction (Exec Time: 0.9ms)
- Task OsTask_20ms_Core1 (Priority 10, Core 1):
  * CruiseControlSWC_MainFunction (Exec Time: 1.2ms)
- Task OsTask_20ms_Core0 (Priority 12, Core 0):
  * HighVoltageMonitorSWC_MainFunction (Exec Time: 0.8ms)
  * CanIf_MainFunction_Tx / CanIf_MainFunction_Rx
- Task OsTask_100ms_Core2 (Priority 5, Core 2):
  * ThermalManagementSWC_MainFunction (Exec Time: 1.5ms)
  * Dem_MainFunction (Exec Time: 2.1ms)
  * Dcm_MainFunction (Exec Time: 1.4ms)

Section 19.0: ECU Mode Management & State Sequences

19.1 ECU State Manager (EcuM) States:
- STARTUP: Phase I (Hardware init), Phase II (OS initialization, NvM restore).
- RUN: Normal operating state with all cyclic tasks executing.
- SHUTDOWN: Prepare NvM block saving, write DTC snapshots, disable CAN transceivers.
- SLEEP: Low-power microcontroller standby with wake-up pin monitoring.

19.2 Wakeup Sources:
- Wakeup Event CAN0_Rx: Vehicle unlock or ignition-on message on CAN bus.
- Wakeup Event KL15_Pin: Physical ignition switch terminal 15 line transition.
- Wakeup Latency: Full operational RUN state achieved within 45ms of signal trigger.

Section 20.0: Glossary, Compliance Declaration & Approvals

20.1 Technical Acronyms and Abbreviations:
- AUTOSAR: Automotive Open System Architecture
- ASIL: Automotive Safety Integrity Level (ISO 26262)
- BSW: Basic Software
- CDD: Complex Device Driver
- DCM: Diagnostic Communication Manager
- DEM: Diagnostic Event Manager
- DTC: Diagnostic Trouble Code
- E2E: End-to-End Signal Protection
- HLD: High-Level Design Document
- MCAL: Microcontroller Abstraction Layer
- NvM: Non-Volatile Memory Manager
- PduR: PDU Router
- RTE: Runtime Environment
- SWC: Software Component
- UDS: Unified Diagnostic Services (ISO 14229)

20.2 Architectural Compliance Statement:
This High-Level Design specification conforms to AUTOSAR Classic Release 4.4.0 standards and ISO 26262:2018 functional safety guidelines for ASIL B system implementations.